SERIEN: An AI-Powered Emotion-Aware Teleconsultation Platform for Remote Mental Health Monitoring

Computer Science and Engineering Department
Academic Year 2025–2026

***Abstract*—Mental health disorders affect one in eight people globally, yet access to professional support remains limited in underserved regions. This paper presents SERIEN, a web-based teleconsultation platform integrating real-time facial emotion recognition with AI-powered clinical insights. The system employs a Convolutional Neural Network (CNN) trained on FER-2013 to classify seven emotions from patient video feeds during WebRTC sessions. Emotion data generates real-time stress indicators and confidence timelines visible to therapists. A LLaMA-based language model integrated through Groq API powers an AI chatbot, generates therapeutic assignments, analyzes journals, and produces post-session reports. The CNN achieves 68.7% validation accuracy on FER-2013 with inference latency under 200 milliseconds. Comparative analysis demonstrates SERIEN's advantages over existing teletherapy solutions through emotion-aware sessions, AI-driven personalization, and continuous patient monitoring.**

***Index Terms*—Teleconsultation, Emotion Recognition, CNN, Deep Learning, Mental Health, WebRTC, LLaMA, Firebase, Real-Time Analysis, Teletherapy**

I. Introduction

The World Health Organization reports depression and anxiety cost the global economy $1 trillion annually in lost productivity. Despite growing burden, the mental health professional-to-patient ratio is critically low in developing nations, with some regions reporting fewer than one psychiatrist per 100,000 population. The COVID-19 pandemic accelerated telehealth adoption, revealing both potential and limitations of remote mental care delivery.

Current platforms (BetterHelp, Talkspace) serve primarily as communication channels without leveraging artificial intelligence for clinical augmentation. A critical limitation is their inability to detect emotional states from facial expressions—information therapists routinely rely upon in-person. SERIEN addresses these gaps through three pillars: (1) a CNN-based facial emotion recognition pipeline running client-side in the therapist's browser using TensorFlow.js, enabling real-time detection without external video transmission; (2) a LLaMA-powered AI engine generating chatbot responses, therapeutic assignments, journal analysis, and clinical reports; (3) a comprehensive workflow encompassing appointments, WebRTC sessions, analytics, journaling, and automated communications.

II. Literature Review

Mollahosseini et al. [1] surveyed deep neural networks for facial emotion recognition (FER), establishing CNN architectures as dominant approaches. The FER-2013 dataset introduced by Goodfellow et al. [2] contains 35,887 grayscale images across seven emotions, remaining the primary FER benchmark with state-of-the-art methods achieving 73–75% accuracy.

Client-side face detection through TensorFlow.js enables real-time browser-based inference without server round-trips. WebRTC for telehealth, examined by Okonkwo and Ade-Ibijola [3], offers end-to-end encryption for patient privacy. Firebase provides serverless infrastructure successfully employed in health informatics applications. LLaMA language models (Touvron et al. [4]) have shown promise in clinical dialogue generation, though integration with emotion-aware teletherapy remains understudied.

III. System Architecture

SERIEN implements a layered hybrid monolith comprising five tiers:

A. Client Tier

React 18 single-page application with Vite build system. Therapist interface loads face-api.js for real-time emotion inference on remote patient video streams. Patient interface supports journaling, appointments, and AI chatbot interaction.

B. Real-time Signaling

Socket.IO manages WebRTC signaling. In-memory session maps track peer connections, forwarding SDP offers, answers, and ICE candidates between peers. Media streams flow directly via WebRTC; no video traverses the server.

C. API and Orchestration

RESTful endpoints expose Groq LLaMA API access for chat completions, assignment generation, email workflows (confirmations, reminders, reports), and journal media upload to Firebase Storage.

D. Data and Identity

Firebase Authentication supports email/password and Google OAuth. Cloud Firestore organizes data across collections: *users*, *sessions*, *reports*, *journals*, *assignments*, *therapist-Patients*, *responses*. Firestore security rules enforce role-based access constraints.

E. Intelligence

CNN emotion recognition runs client-side; LLaMA accessed server-side through Groq API generates chat, assignments, and reports.

IV. CNN Emotion Detection Model

A. Architecture

The model processes 48×48 grayscale facial images through sequential layers:

- 4 Conv2D blocks (128, 256, 512, 512 filters) with 3×3 kernels and ReLU activation
- MaxPooling2D (2×2) after each convolutional layer
- Dropout (0.3–0.4) for regularization
- Flatten layer into 512 dense units
- Output layer with 7 softmax units (emotion classes)

B. Training

Trained on FER-2013 (28,709 training, 7,178 validation images) using Adam optimizer (lr=0.0001), categorical cross-entropy loss, batch size 64, 50 epochs. Data augmentation includes random flipping, rotation (±10°), and zoom. Model weights stored in HDF5 format (48.5 MB); architecture serialized as JSON for TensorFlow.js deployment.

C. Confidence Scoring

Softmax output produces probability distribution across seven emotions. Confidence score corresponds to highest probability. Stress score computed as weighted combination: $\textit{stress} = 0.4 \times \textit{fearful} + 0.35 \times \textit{angry} + 0.25 \times sad.Livealertstriggerwhenstressexceedsconfigurablethreshold.$

V. LLaMA AI Integration

LLaMA 3.1 8B model accessed through Groq API provides:

Chatbot: Role-aware system prompts (empathetic for patients, analytical for therapists). When emotion data available, prompt augmented with detected emotion for contextual responses. Temperature 0.6, max 180 tokens.

Assignments: AI-crafted self-reflection MCQ questions focused on emotional awareness, stress triggers, coping behavior. Generates JSON with four options per question, validated before Firestore storage.

Journal Analysis: Identifies emotional patterns, risk indicators, and areas of progress from patient entries.

Report Generation: Post-session reports computed from emotion timeline and AI-generated recommendations with dominant emotions, stress/risk labels, emotion percentages, and follow-up guidance.

VI. Experimental Results

A. Model Performance

CNN validation accuracy: **68.7%** on FER-2013 test set. Per-class precision, recall, F1-score:

Emotion	Precision	Recall	F1
Happy	0.82	0.79	0.81
Sad	0.71	0.68	0.69
Angry	0.75	0.73	0.74
Fear	0.65	0.62	0.63
Neutral	0.73	0.76	0.74
Surprise	0.68	0.65	0.66
Disgust	0.61	0.58	0.59

B. Real-time Performance

Inference latency: **140–200 ms** per frame using face-api.js on standard laptop hardware. Sampling interval: 200 ms, enabling emotion timeline granularity sufficient for therapist monitoring.

C. Comparative Analysis

Feature	SERIEN	BetterHelp	Talkspace
Real-time Emotion Detection	✓	55	55
AI-Powered Reports	✓	55	Partial
Live Stress Alerts	✓	55	55
Journaling	✓	Limited	✓
AI Chatbot	✓	55	55

VII. Conclusion

SERIEN integrates facial emotion recognition, LLaMA-based AI, and comprehensive patient management into a unified teleconsultation platform. The CNN achieves competitive accuracy on FER-2013 with real-time browser-based inference, enabling therapists to observe emotional states during remote sessions. AI-driven assignment generation, journal analysis, and post-session reports enhance clinical decision-making and patient engagement between appointments. The platform demonstrates clear advantages over existing teletherapy solutions through emotion-aware capabilities and continuous monitoring. Future work includes expanding to additional emotion datasets, optimizing inference latency below 100ms, and conducting clinical validation studies with licensed therapists and actual patients in therapeutic settings.

References

[1] A. Mollahosseini, B. Hasani, and M. H. Mahoor, "Affectnet: A database for facial expression, face recognition, and emotion recognition," *IEEE Trans. Affective Comput.*, vol. 10, no. 1, pp. 18–31, 2017.

[2] I. J. Goodfellow, D. Erhan, D. Dumoulin, and Y. Bengio, "Challenges in representation learning: A report on three machine learning contests," arXiv preprint arXiv:1307.0414, 2013.

[3] C. W. Okonkwo and A. Ade-Ibijola, "Chatbots applications in education: A systematic review," *Comput. Educ. Artif. Intell.*, vol. 2, p. 100033, 2021.

[4] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, and S. Batra, "Llama 2: Open foundation and fine-tuned chat models," arXiv preprint arXiv:2307.09288, 2023.

[5] World Health Organization, "Mental health and psychosocial support," https://www.who.int/teams/mental-health-and-substance-use, 2021.